LABORATORY

4

Shedding Light on the Structure of the Atom

Objectives

- Build and calibrate a simple spectroscope.
- Use your spectroscope to observe the line spectrum of sodium; calculate the energy and frequency associated with the major transition(s).
- Use your spectroscope to observe the line spectra of unknown substances; use line spectrum to identify substance.

Materials Needed

- A one-piece box (8" × 8" × 2") [versions can be purchased at The Container Store (item 162004), or online from various sources including FetPak Inc. (http://www.fetpak.com; item AW22) and Paper Mart (http://www.papermart.com; item 4011480)]
- Holographic diffraction grating (slide, or 24 mm × 36 mm piece of paper); 1000 lines/mm preferred [multiple sources online include Rainbow Symphony Store (http://www.rainbowsymphonystore.com; item 01504 or 01603)]
- Aluminum foil to make slit
- Black tape (duct, electrical)
- Scotch tape
- Ruler
- Scissors
- Something to cut box (e.g., razor blade, box cutter, or exacto knife)
- Light sources for observation (e.g., gas discharge tubes filled with various gases)

Safety Precautions

- **Never look directly at sunlight or a laser light (e.g., a laser pointer) with your spectroscope!** If you are using one of these as your light source, look at it indirectly [as it reflects off of a white surface (e.g., wall, sheet of paper, etc.)].
- Some light sources (e.g., hydrogen and nitrogen lamps) may emit ultraviolet radiation, which can damage your eyes. The typical safety glasses will protect your eyes from the UV radiation, so **keep your safety glasses on at all times.**
- If you are working with gas discharge tubes, be sure to unplug the power supply before removing or adjusting the position of the tube.
- If you are working with open flames as source of light, be sure not to catch your spectroscope on fire.

Background

Because atoms are too small to be observed directly, scientists in the late nineteenth and early twentieth centuries had to use indirect methods to decipher the architecture of individual atoms. These classic and creative experiments, which included cathode ray tubes, Millikan's oil drop experiment, and Rutherford's gold foil experiment, led to the atomic model with a nucleus containing protons and neutrons surrounded by a relatively large "cloud" of electrons.

But are the electrons further organized around the nucleus, or is the implied randomness in the preceding "cloud" analogy accurate? Again, scientists have been forced to utilize indirect methods to understand how electrons are arranged in an atom. For this question, **light** has been utilized to probe the electronic structure of atoms. Specifically, scientists have used light (or more accurately electromagnetic radiation), and the absorption and emission of light by matter, to determine that indeed electrons *do* have a specific organization in each and every atom.

The classic (and most simple) example for us to consider is hydrogen, which has one proton in the nucleus and one electron. When hydrogen gas is sealed inside a glass tube and submitted to an electric current, we observe a reddish glow emanating from the tube. [Note: Light is emitted when any matter absorbs enough energy: hot objects generate light (e.g., piece of metal in a fire), gases emit light when an electric current is passed through them, and some forms of matter will burn with a noticeable color when placed in a flame.] When we pass this light through a slit and a prism or a diffraction grating (see the previous lab), we observe a series of individual lines of different colors and *not* a continuous spectrum with all wavelengths present. Interestingly, we *always* see the exact same lines for hydrogen—never more or less, and never at different wavelengths (colors in the visible region of the electromagnetic spectrum). This series of lines is called an atomic spectrum or line spectrum, and it is diagnostic for the presence of hydrogen; in other words, different elements will generate different line spectra (each unique for that specific element). The lines for hydrogen that are observed in the visible region have been well characterized (see Table 1 and Figure 1).

Table 1 Line Spectrum Values for Hydrogen in the Visible Region.[1]

Wavelength (nm)	Color
383.5384	Violet
388.9049	Violet
397.0072	Violet
410.174	Violet
434.047	Violet
486.133	Blue-green
656.272	Red
656.2852	Red

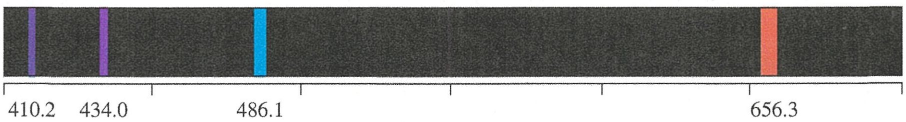

Figure 1 Line spectrum for hydrogen.

This observed spectrum generates several questions:

- How are these lines generated?
- How can there be multiple lines if hydrogen only has one electron?
- Why are these lines at these specific wavelength values? Why not different wavelength values? Why not more or fewer lines?
- Why are line spectra different for different elements?

Most importantly, any successful model of the atom must provide an understanding and explanation of the line spectral data.

Based on the work of Niels Bohr,[2-4] we have a simple model for the atomic structure of the atom that can help explain the line spectrum of hydrogen. In this model, electrons (e^-) in an atom are located in fixed orbits that encircle the nucleus. An e^- in a given orbit has a specific energy given by Equation 1:

$$E = -\frac{2\pi^2 me^4}{n^2h^2} = -\frac{b}{n^2} \qquad \textbf{Equation 1}$$

where m is the e^- mass, e is the e^- charge (-1.6022×10^{-19} C), h is Planck's constant (6.63×10^{-34} J s), and $b = 2.178 \times 10^{-18}$ J (combination of all constants in the equation). The variable n can have values 1, 2, . . . and is known as the **principal quantum number.** Each orbit has a different value of n; the lowest orbit (closest to the nucleus) has $n = 1$, the next farthest from the nucleus has $n = 2$, and so forth. An e^- located in the first Bohr orbit ($n = 1$) has the lowest (most negative) energy because it is closest to the nucleus. As the value of n increases, an e^- will be located farther from the nucleus and therefore have a higher (more positive) energy. Another important tenet of Bohr's model is that the e^- can only have an energy associated with one of the orbits; that is, the e^- can never have an intermediate energy. Thus, we say that the energy of an e^- is **quantized.**

Using this model, hydrogen has one e^- located in the $n = 1$ Bohr orbit. This e^- is able to move away from the nucleus by **absorbing** a specific amount of energy and relocating to a new orbit with a larger n value (position of higher potential energy)—this is known as an **excited state** of the atom, and it is less stable than the original **ground state** (lowest energy possible). The difference between the energies of the two orbits between which the e^- moves reflects the amount of energy absorbed by the e^-. This difference can be calculated using Equation 2:

$$\Delta E = E_f - E_i = -b\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right) \qquad \textbf{Equation 2}$$

Here the subscripts i and f stand for initial and final states of the e^-, respectively. Being unstable, the e^- will eventually release a specific amount of energy in the form of light **(emission)** and return to a lower energy orbit, with ΔE calculated using Equation 2. Bohr reasoned that the lines observed in the atomic spectrum of hydrogen arise from the e^- transitions between the different orbits. Since the orbit energies are quantized, ΔE can have only specific values, and therefore only specific lines are observed with specific wavelengths (colors) associated with them.

$$c = \lambda\nu \qquad \Delta E = h\nu \qquad \Delta E = (hc)/\lambda \qquad \textbf{Equations 3a–c}$$

where

c = speed of light = 2.998×10^8 m s^{-1}

h = Planck's constant = 6.63×10^{-34} J s (photon)$^{-1}$

ν = frequency (s^{-1} or Hz)

λ = wavelength (m)

Although the simple Bohr model works only for hydrogen and any other one e^- ion (the model is incomplete for multielectron systems because it does not account for interactions between the electrons), we can still use Equations 3a–c to relate the wavelength of a given line in a line spectrum of any element to the ΔE that an e^- undergoes.

The line spectrum of hydrogen has been heavily studied for a number of reasons. With only one electron involved, it should be the easiest to interpret. Also, it is useful in astronomy, since these lines show up in the spectra associated with stellar objects in the universe that are commonly studied based on their emitted light. The Balmer series for hydrogen[5] is the series of lines that result from the e^- relaxing from various final orbits (n_f) to the $n = 2$ orbit. We can modify Table 1 to include the orbit-to-orbit transitions that are specifically involved for each line, generating Table 2.

Table 2 Balmer Series for Hydrogen in the Visible Region.[5]

Wavelength (nm)	Color	Transition ($n_f \rightarrow n_i$)
383.5384	Violet	$9 \rightarrow 2$
388.9049	Violet	$8 \rightarrow 2$
397.0072	Violet	$7 \rightarrow 2$
410.174	Violet	$6 \rightarrow 2$
434.047	Violet	$5 \rightarrow 2$
486.133	Blue-green	$4 \rightarrow 2$
656.272	Red	$3 \rightarrow 2$
656.2852	Red	$3 \rightarrow 2$

In this lab, the goal is to create a spectroscope that will use a diffraction grating to separate light emission into a rainbow (for a continuous light source) or a line spectrum for elemental sources; numerous variations on the spectroscope design used here can be found in References 6–12. You will then calibrate the spectroscope using the fluorescent lights in the lab as your light source. Finally, you will use the spectroscope to measure the transition energies for various elements as well as to measure line spectra for unknown samples to determine their identities.

References

1. *CRC Handbook of Chemistry and Physics.* 53rd ed. (Weast RC, Ed.). 1972; The Chemical Rubber Co, Ohio.
2. Bohr N. On the Constitution of Atoms and Molecules, Part I. *Philosophical Magazine.* 1913; 26: 1–24.
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5. Birge RT. The Balmer Series of Hydrogen, and the Quantum Theory of Line Spectra. *Phys. Rev.* 1921; 17: 589–607.
6. Edwards RK, Brandt WW, Companion AL. A Simple and Inexpensive Student Spectroscope. *J. Chem. Ed.* 1962; 39: 147–148.

7. Wakabayashi F. Resolving Spectral Lines with a Periscope-Type DVD Spectroscope. *J. Chem. Ed.* 2008; 85: 849–853.
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9. Cortel A, Fernandez L. A Simple Diffraction Grating Spectroscope: Its Construction and Uses. *J. Chem. Ed.* 1986; 63: 348–349.
10. Westra MT. A Fresh Look at Light: Build Your Own Spectrometer. *Science in School.* 2007: 30–34.
11. Briggs RP, Carlisle RJ. *Building a Spectroscope.* In *Solar Physics and Terrestrial Effects: A Curriculum Guide for Teachers Grades 7–12.* 2nd ed. (Poppe BB, Ed.), pp. 104. 1996; Space Environment Center, National Oceanic and Atmospheric Administration.
12. Website www.swpc.noaa.gov/Curric_7-12/Activity_1.pdf (last accessed July, 2013).

Procedure

I. Construct Spectroscope

1. Fold an 8 in × 8 in × 3.5 in (203 mm × 203 mm × 89 mm) box.
2. Use a strip of black tape on the inside of the box to secure the bottom fold and prevent light from leaking through.
3. On the bottom face of the box, use your ruler to draw a line that splits the box bottom in half; this is the long leg of a right triangle (connects points A and B in Figure 2).
4. To generate the correct angle (15°) between the slit and the diffraction grating, the short leg of the triangle (line B–C in Figure 2) will measure 53 mm long. Note that the length of leg B–C will be different if you use a box that has a length (leg A–B) different than 8 in; to calculate, use Equation 4.

$$\tan(\theta) = \frac{B - C}{A - B} \qquad \textbf{Equation 4}$$

Here θ = angle C–A–B = 15°, and A to B and B to C are the lengths of the two legs of the triangle (Figure 2).

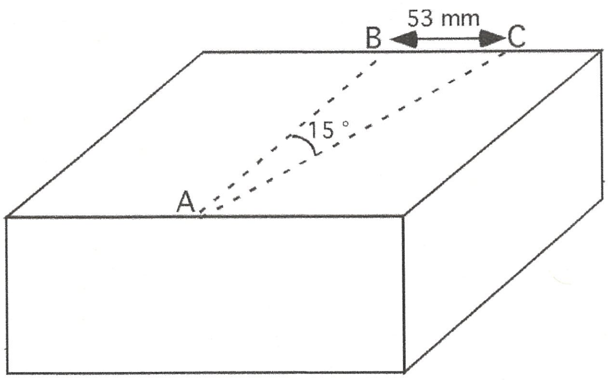


Figure 2

5. Incorporate the slit (Figure 3):
   - Cut a 20 mm (wide) × 40 mm (high) rectangular hole centered above point C.
   - Cut a piece of aluminum foil in half.
   - Crease each piece to generate a sharp edge on each.
   - Tape the two pieces of foil next to each other over the hole in the box to create a narrow vertical slit.

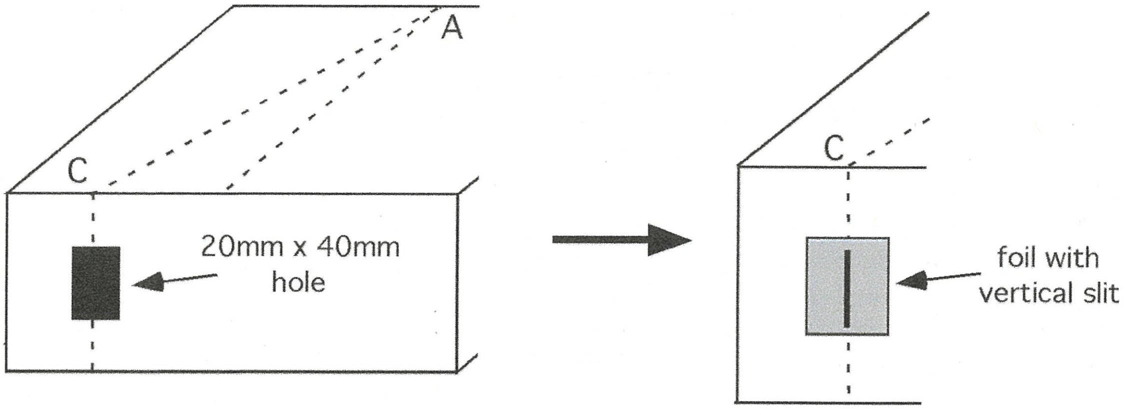


Figure 3

6. Incorporate the diffraction grating (Figure 4):
    - On the end of the box above point A, cut a 34 mm (wide) × 24 mm (high) rectangular hole, centered above A.
    - Take a piece of diffraction grating paper 50 mm (wide) × 40 mm (high), put a piece of Scotch tape on one edge.
    - On the inside of the box, place the grating over the hole, square with the sides of the hole.
    - Close the box, point the slit directly at the fluorescent lights in the room, and look for a line spectrum to the right of the slit inside the box.
    - If a spectrum does not appear, then your grating needs to be rotated 90° to align the grating lines with the slit orientation.
    - Once properly aligned, use Scotch tape to tape down the other three edges of the grating.

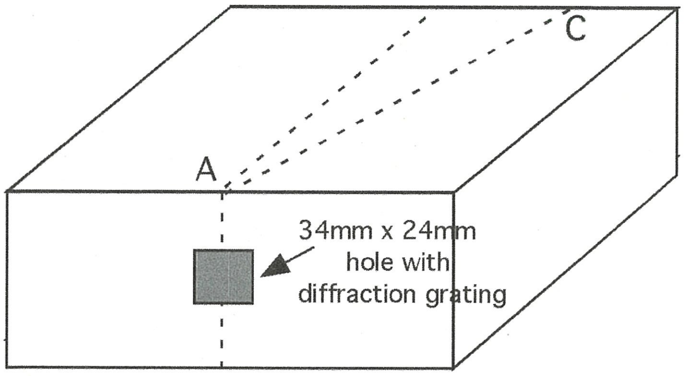
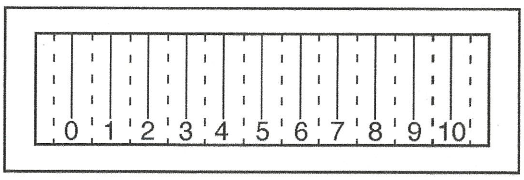


Figure 4

7. Incorporate the grid for spectral measurements:
    - Use a pencil to mark inside the box approximately where the ends of the spectrum appear.
    - Use a razor blade to cut a flap that overlaps this location but is displaced toward the bottom of the box (see the illustration for the approximate configuration); the flap should be hinged at the *top* (toward the middle of the box), and should open from the *bottom*; the goal is to have the hole that you've cut slightly overlap the line spectrum; the hole should be ~60 mm (wide) × ~15 mm (high).
    - Copy this grid from lab manual:



- Use Scotch tape to attach the grid over the hole on the inside of the box; the numbers and lines should point to the inside of the box (Figure 5).

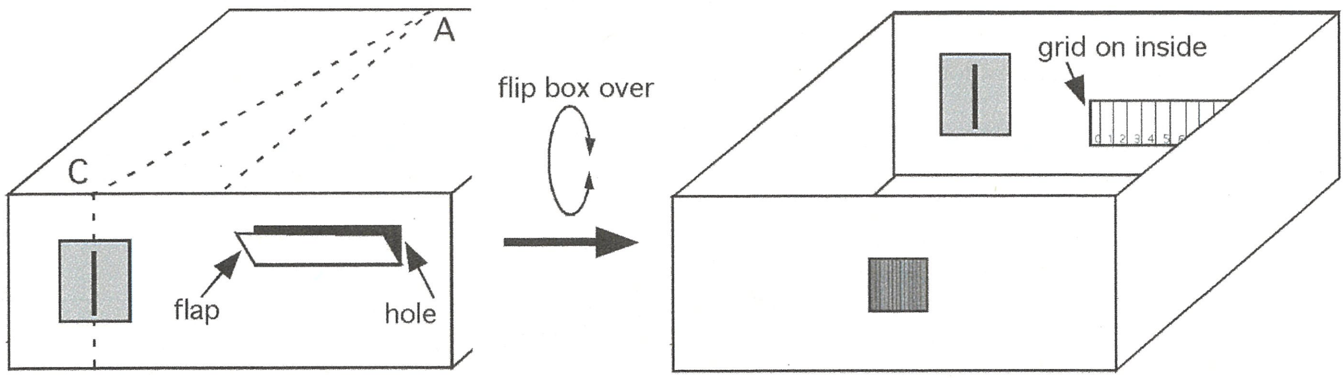


Figure 5

8. You can now close the box and, if too much light leaks in, seal the box shut using black tape. The goal is to have the slit, the diffraction grating, and the measurement grid window be the only sources of light inside the box.

II. Calibrate Spectroscope

To make the spectroscope useful as a tool for quantitative measurements, we must calibrate it using a light source that generates a well-characterized line spectrum with precisely known wavelengths. The fluorescent lights in your lab contain mercury vapor and work well for calibration purposes. The known wavelength values for observed lines in the Hg line spectrum are listed in Table 3.[1]

Table 3 Line Spectrum Values for Mercury in the Visible Region.[1]

Color	Wavelength (nm)
Violet	404.7
Blue	435.8*
Green	546.1*
Yellow-orange	577.0*
Yellow-orange	579.1*
Orange-red	623.4
Red	690.7

* Most intense lines

1. Point the slit of your spectroscope toward the fluorescent lights in the lab while looking through the diffraction grating; a line spectrum should appear to the right of the slit.
2. Adjust the grid opening so that you can see both the line spectrum and the grid; for each line, record the color and its position on the numbered grid on the data sheet.
3. Use the grid position versus known wavelength data to generate a best-fit (regression) line for use as a calibration curve (see the Analysis section).

III. Measurements on Sodium Sample and Unknown Sample(s)

1. Using the procedure just described, record the color and position of each line for the sodium sample. Sodium has a prominent line called the sodium D line. This line is due to the valence electron in Na moving from an excited state (a 3*p* orbital) back to its ground state 3*s* orbital.
2. Record the color and position of each line for each unknown sample.

Data Collection

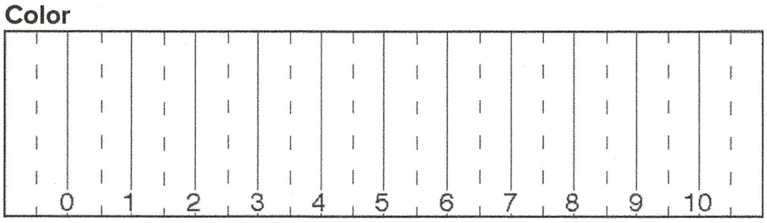
1. Calibration data for spectroscope:
    - Record grid positions in this table.

Color	Wavelength (nm)	Grid Position
Violet	404.7	
Blue	435.8*	
Green	546.1*	
Yellow-orange	577.0*	
Yellow-orange	579.1*	
Orange-red	623.4	
Red	690.7	

* Most intense lines

2. Line spectrum data for sodium sample:
    - Draw a line on the grid below in the correct position for each observed line.
    - Above each line, indicate the color of the line.
    - Below each line, indicate the grid position numerical value.

Color



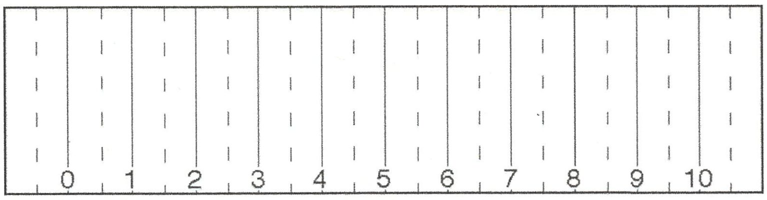
Grid position

3. Line spectrum data for unknown sample(s): for each sample,
    - Draw a line on the grid below in the correct position for each observed line.
    - Above each line, indicate the color of the line.

- Below each line, indicate the grid position numerical value (e.g., if the line is between 2 and 3 on the grid, you might write "2.4" for the grid position).

SAMPLE 1

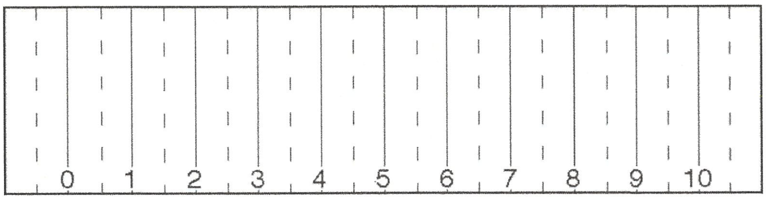
Color



Grid position

SAMPLE 2

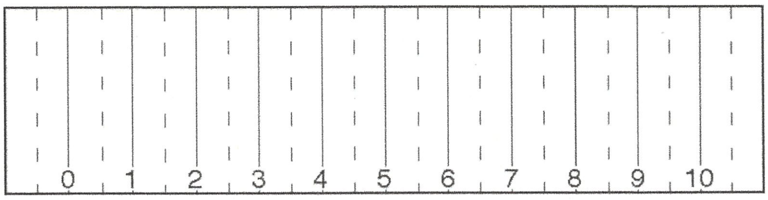
Color



Grid position

SAMPLE 3

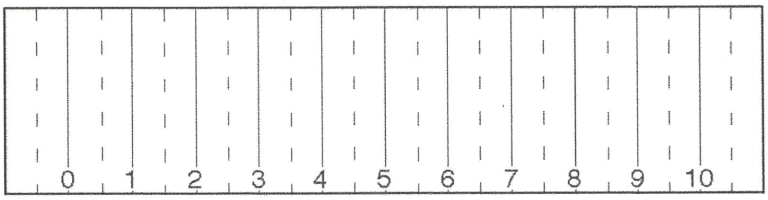
Color



Grid position

SAMPLE 4

Color



Grid position

Analysis

1. Generation of calibration curve:
   - To generate the line that best fits our calibration data, we will be using the method of least squares. The basic idea is to minimize the distance of the line from all the data points concurrently. The calculus behind this method (which involves partial derivatives) is beyond the scope of this lab, yet the calculations are straightforward enough.

- The line $y = mx + b$, the least-squares or best-fit line that is associated with the data points $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$, can be calculated using the following equations:

slope: $$m = \frac{n(\sum xy) - (\sum x)(\sum y)}{n(\sum x^2) - (\sum x)^2}$$ **Equation 5**

y-intercept: $$b = \frac{\sum y - m(\sum x)}{n}$$ **Equation 6**

where

$$\sum xy = x_1y_1 + x_2y_2 + \cdots + x_ny_n$$
$$\sum x = x_1 + x_2 + \cdots + x_n$$
$$\sum y = y_1 + y_2 + \cdots + y_n$$
$$\sum x^2 = x_1^2 + x_2^2 + \cdots + x_n^2$$

a. Fill in the following table based on the calibration data you collected:

x (wavelength)	*y* (grid position)	*xy*	x^2
404.7			
435.8*			
546.1*			
577.0*			
579.1*			
623.4			
690.7			

* Most intense lines

b. Use the values from part (a) to calculate the following values:

$\sum x$ = ______________

$\sum y$ = ______________

$\sum xy$ = ______________

$\sum x^2$ = ______________

c. Plug the values from part (b) into Equations 5 and 6 to calculate the slope and y-intercept of the least squares line [n is the number of (x, y) data points from part (a)].

m = ______________

b = ______________

d. Using your data in part (c), plot your line spectrum data and least-squares line on the following graph, or plot your data with Excel, get the best fit line and put a printout of your graph with the best fit line here:

2. Calculate the energy associated with the line from the sodium sample:
   a. Use your calibration curve to determine the wavelength of the sodium line you observed. Show your calculations:

      $\lambda =$ ____________________

   b. The literature value for the wavelength associated with the sodium line is 589 nm. Calculate the percent error associated with the wavelength you calculated, where the percent error is calculated using Equation 7:

$$\% \text{ error} = \left| \frac{(\text{measured}) - (\text{literature})}{\text{literature}} \right| \times 100 \qquad \textbf{Equation 7}$$

      % error = ____________________

   c. Convert the wavelength you generated in part (a) to the energy associated with this transition. Show your calculations:

      energy = ____________________

3. Identification of unknowns using line spectra:
   - For each unknown, compare the line spectrum you recorded above to the set of known line spectra in Figure 6 to identify each substance:

Substance	Identity (based on line spectra)
SAMPLE 1	
SAMPLE 2	
SAMPLE 3	
SAMPLE 4	

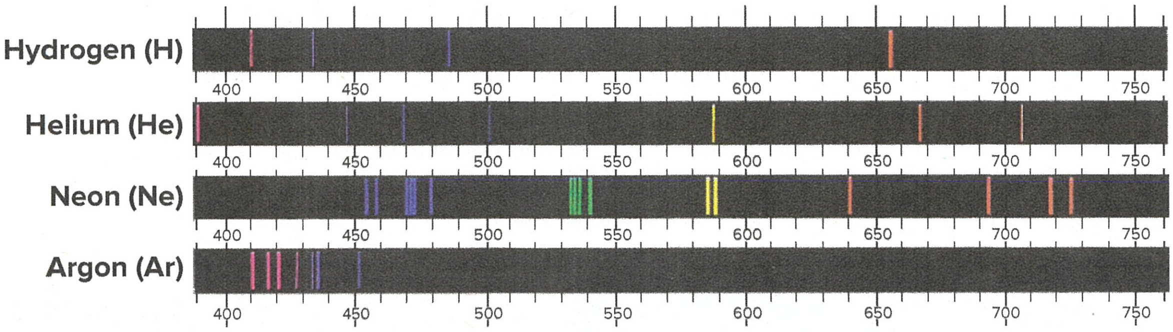


Figure 6 Line spectra of various elements.

Reflection Questions

1. The spectral lines observed for hydrogen arise from transitions from excited states back to the $n = 2$ principal quantum level. Calculate the wavelengths associated with the spectral transitions of the hydrogen atom from the $n = 6, 5, 4$, and 3 levels to the $n = 2$ level.

2. Suppose you use a diffraction grating that contains 600 lines mm^{-1} in your spectroscope. What effect would you expect this to have on the sharpness and resolution of the lines in your line spectrum?

3. Let's imagine that you were going to start a company that produces spectroscopes to sell to general chemistry students. How could you ensure that all the spectroscopes created the same calibration line and equation? What would you need to be sure was always the same? What could be changed without affecting the calibration?

4. What are the possible sources of error in the wavelength of the sodium line you calculated in Analysis part 2(b)?

5. As indicated earlier, white light contains a continuous spectrum including all the visible wavelengths of light. Interestingly, if one uses a spectroscope to look at sunlight (**not directly at the sun, which would cause damage to your eyesight!!**, but indirectly by pointing the spectroscope at a clear part of the sky), we observe numerous dark lines superimposed on the continuous spectrum. This phenomenon was originally observed by Wollaton in 1802 and then again later by Fraunhofer in 1817. Called Fraunhofer lines, these lines correlate with the line spectra of various elements that are thought to exist in the outer atmosphere of the sun (Figure 7).

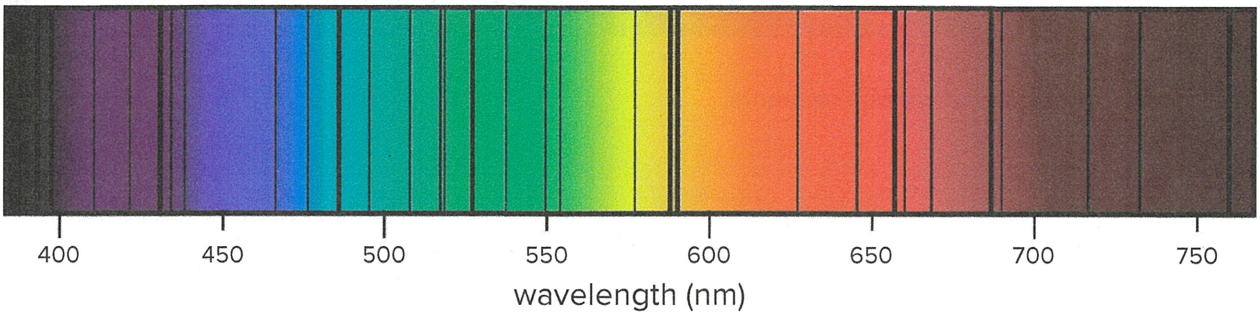


Figure 7 **Fraunhofer lines.**

a. Explain the process by which these dark lines are generated by the elements in the sun's atmosphere. Specifically, why are they black and not colored, and why do they match the colored line spectra of those elements?

b. Compare the Fraunhofer lines to the line spectra in Figure 6. Which of these elements are in the sun?

Connection

Based on your experience in this lab, draw a connection to something in your everyday life or the world around you (something not mentioned in the background section).